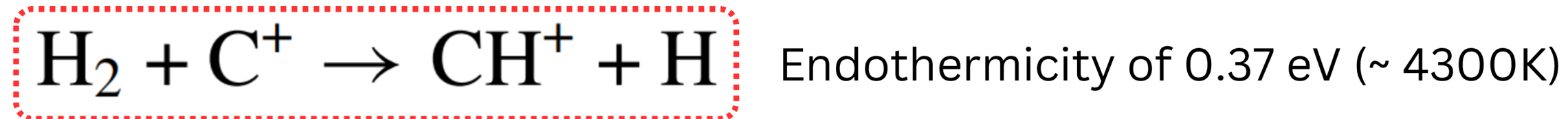


Why study CH⁺ in ISM?

1. How CH⁺ generation takes place in ISM...

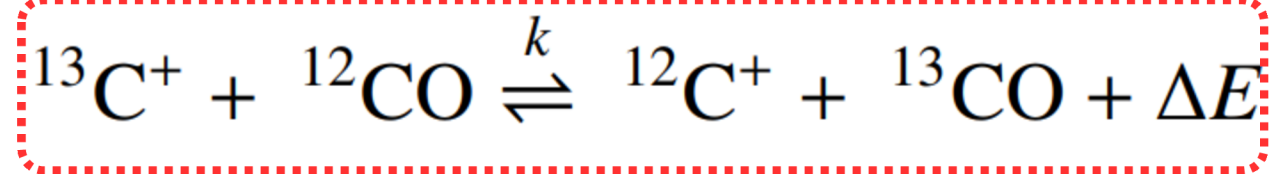


- The sheer abundance of CH⁺ in the interstellar medium inspite of the fact that H₂ is found in the colder parts of the ISM and this reaction needs high temperatures made it a really interesting species to work with.
- Previously mechanisms like shock heating, Alfven waves and also vibrationally active H₂ by FUV radiation have been considered.
- More recently, it has been found that CH⁺ is mostly formed on the **boundary of warm (C⁺) and cold regions (H₂) of the ISM by turbulence dissipation and thermal instability** ([Godarad, et al. 2023](#))

Why study CH⁺ in ISM?

2. The isotopologue ratio ¹²CH⁺/¹³CH⁺ is similar to ambient ¹²C/¹³C ratio...

- It does not participate in chemical fractionation (which enriches molecules in ¹³C and thus lowers the ratio). Mostly formed in the edges of molecular clouds where temperatures are much higher than what fractionation needs.

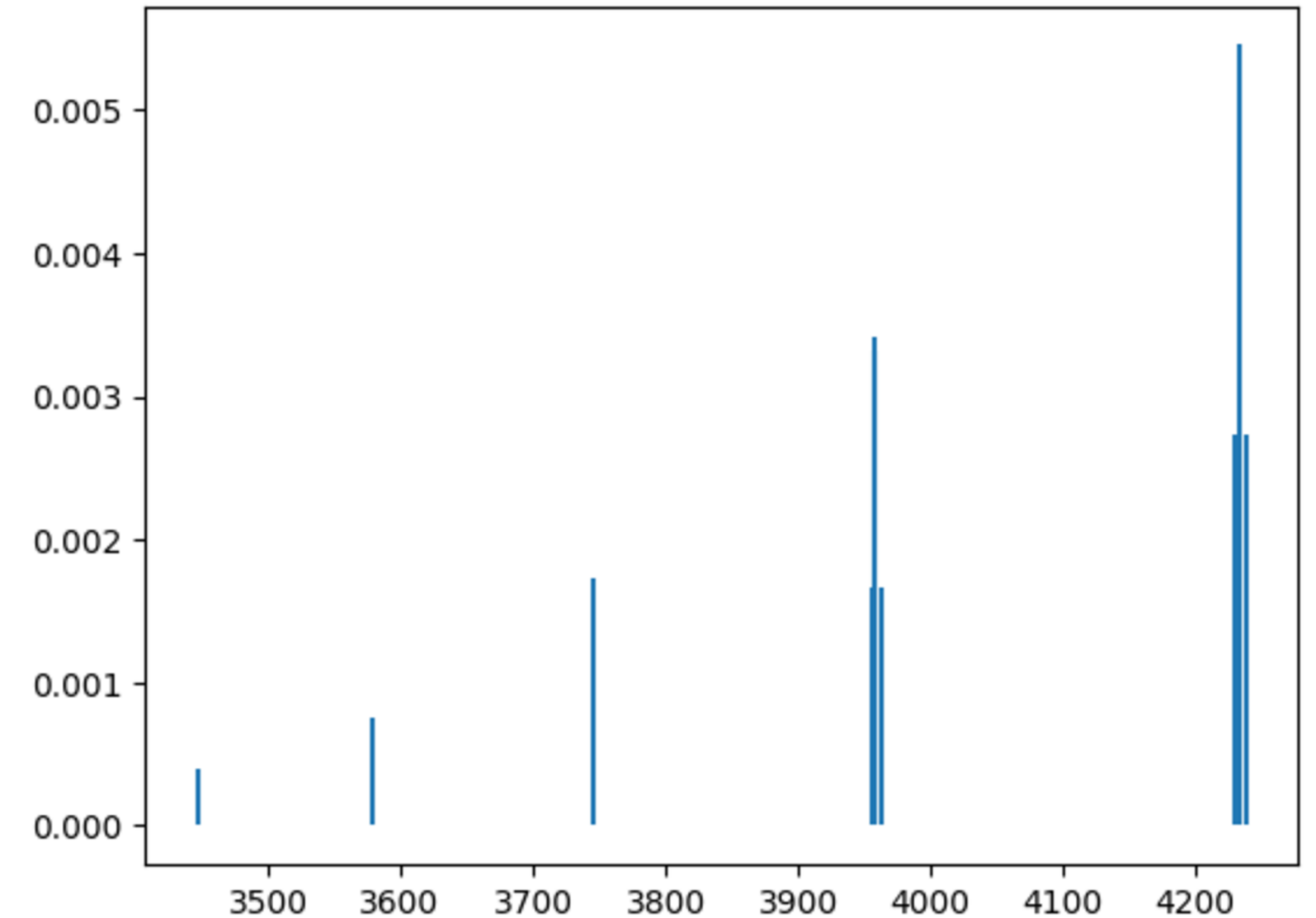


$$\frac{n(^{13}\text{C}^+) n(^{12}\text{CO})}{n(^{12}\text{C}^+) n(^{13}\text{CO})} = \exp\left(-\frac{\Delta E}{kT}\right) \quad (\text{at equilibrium})$$

$$\Delta E/k = 35 \text{ K and } k = 2 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$$

- It does not undergo selective photodissociation ([Casassus, et al.](#))

Species	Wavelength (Å)	Oscillator strength
$^{12}\text{CH}^+$	3447.077	0.000400
$^{12}\text{CH}^+$	3579.024	0.000750
$^{12}\text{CH}^+$	3745.308	0.001720
$^{12}\text{CH}^+$	3955.486	0.001655
$^{12}\text{CH}^+$	3957.689	0.003420
$^{13}\text{CH}^+$	3958.129	0.003420
$^{12}\text{CH}^+$	3962.071	0.001655
$^{13}\text{CH}^+$	4232.288	0.005450
$^{12}\text{CH}^+$	4232.548	0.005450
$^{12}\text{CH}^+$	4229.347	0.002725
$^{12}\text{CH}^+$	4237.560	0.002725



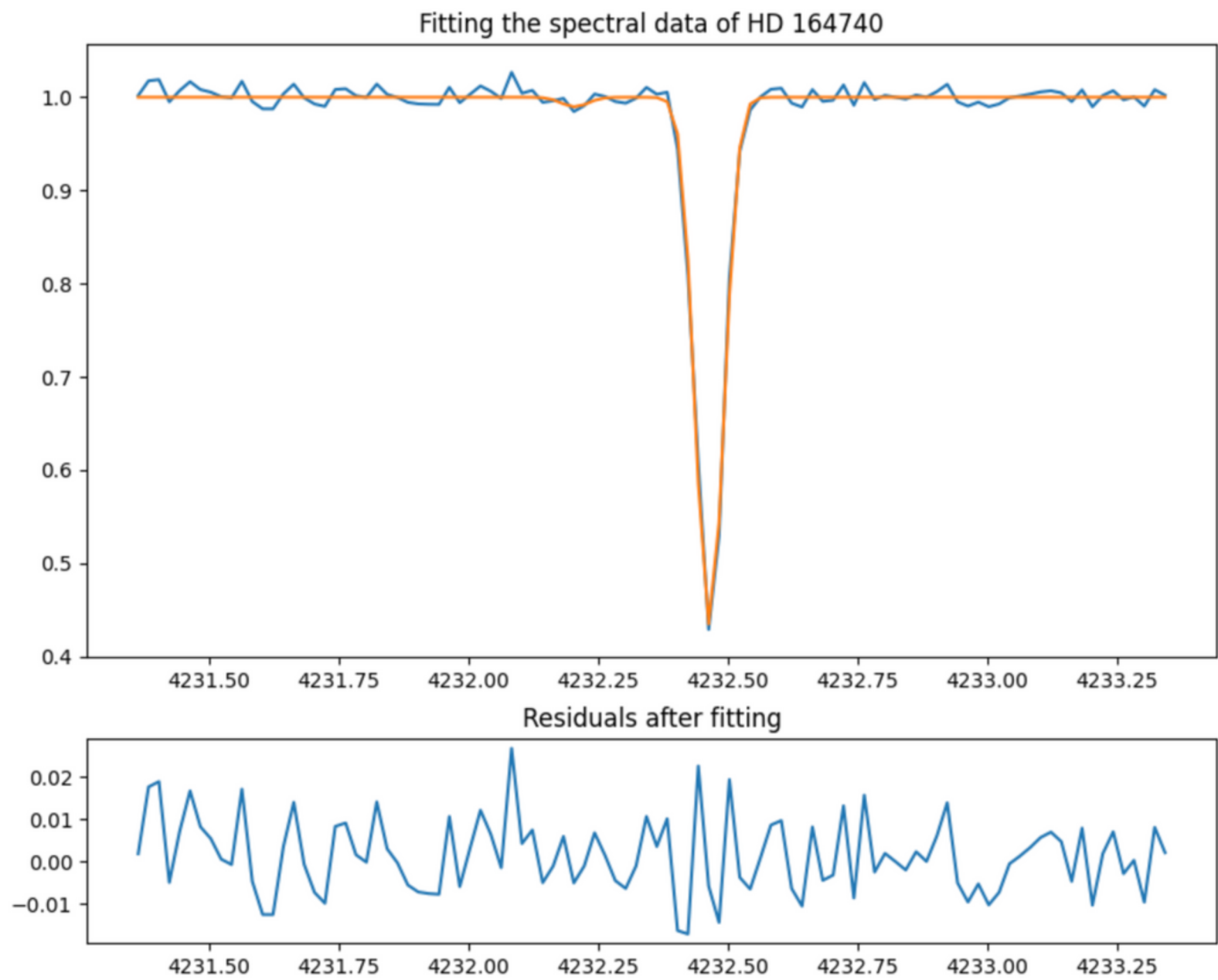
- For this work, we focussed on the strongest CH^+ absorption line at 4232 Å, which corresponds to the R(0) rotational line of the **$\text{A}1\Pi \rightarrow \text{X}1\Sigma^+ (0, 0)$** vibronic band and has an oscillator strength of 0.00545.
- Found that >80% of all the EDIBLES targets have CH^+ in their sightlines. Probably because most of these sightlines are along the boundaries of cold and warm regions of the ISM.

A small subset of the results (single cloud models)

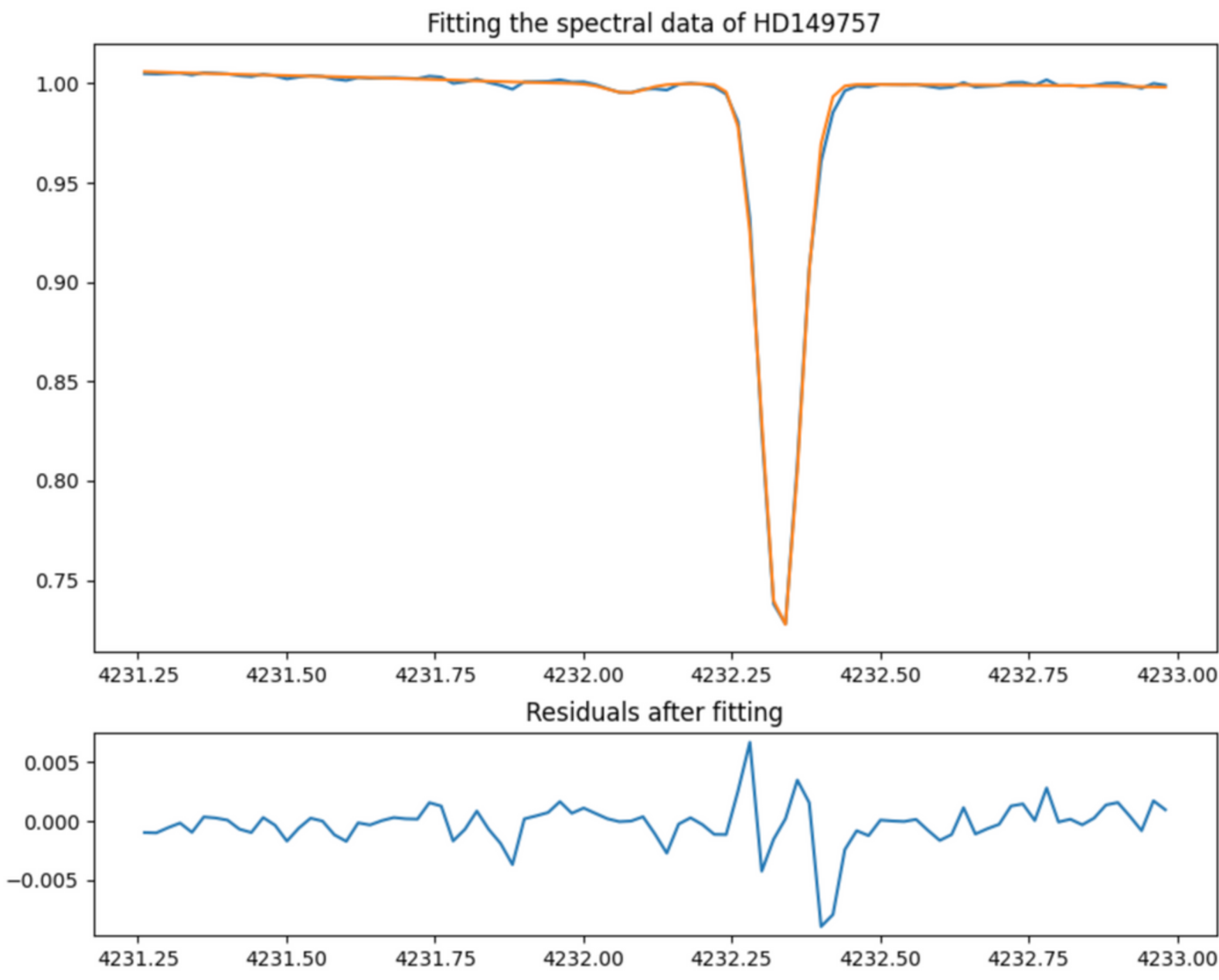
Sightline	N	b	v	R	N_{lit}	b_{lit}	v_{lit}	R_{lit}	Reference
HD55879	0.4	2.9 ± 0.1	23.0 ± 0.1	..	0.4 ± 0.1	..	23.8	..	1
HD149757	3.00 ± 0.02	2.55 ± 0.02	-15.23 ± 0.01	72 ± 19	3.2 ± 0.1	2.0	-19.4	75 ± 12	2
HD110432	1.70 ± 0.01	2.62 ± 0.02	5.26 ± 0.02	72 ± 24	1.7 ± 0.2	..	..	62 ± 5	3, 4
HD37367	3.34 ± 0.02	2.51 ± 0.03	16.03 ± 0.01	114 ± 61	3.2	..	..	..	5
HD54439	1.91 ± 0.02	2.6	32.06 ± 0.02	40 ± 10	..	..	..	..	..
HD73882	2.19 ± 0.02	3.6 ± 0.1	20.86 ± 0.03	64 ± 30	2.4 ± 0.3	21.5	..	..	1
HD164740	6.7 ± 0.1	1.79 ± 0.05	-5.92 ± 0.03	87 ± 51	..	..	..	..	..
HD164906	1.18 ± 0.02	2.74 ± 0.06	-6.04 ± 0.04	36 ± 14	..	..	..	..	..

Table 1. Comparison of CH^+ column density (N), Doppler parameter (b), and velocity (v) with literature values. Column densities are in units of 10^{13}cm^{-2} , b and v in km/s . References: [1: Gredel 2002], [2: Ritchey 2010], [3: Casassus 2005], [4: Gredel 1992], [5: Sheffer 2008]

A small subset of the results (single cloud models)



Bad 13CH+ fit

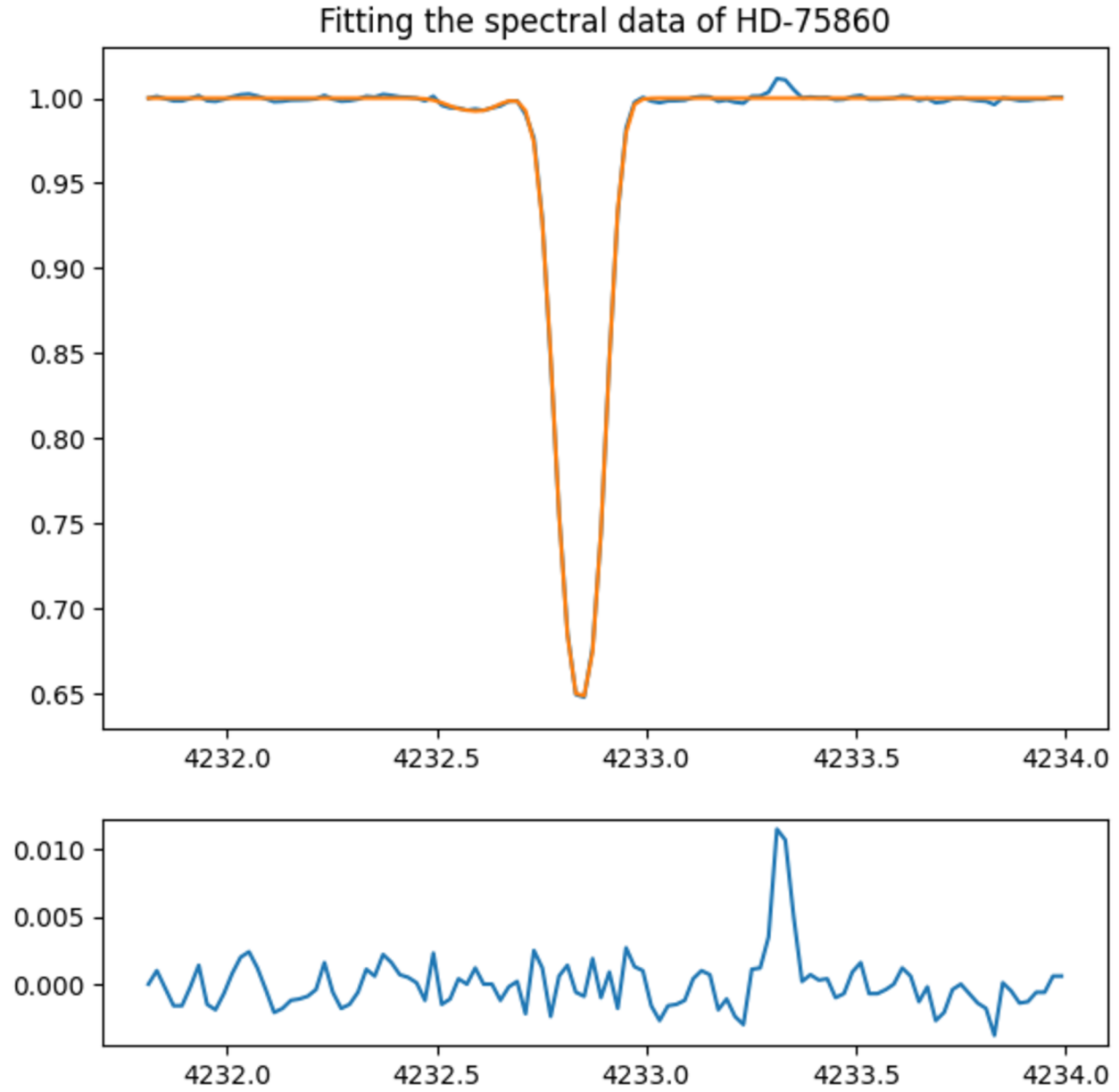


Good 13CH+ fit

A small subset of the results (double cloud models)

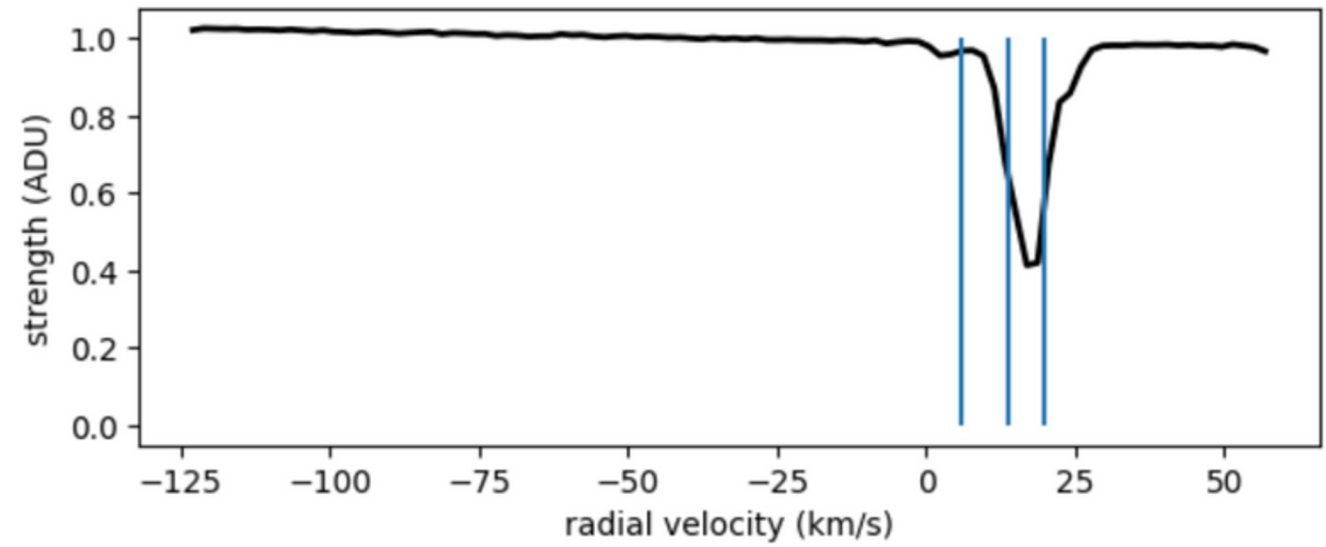
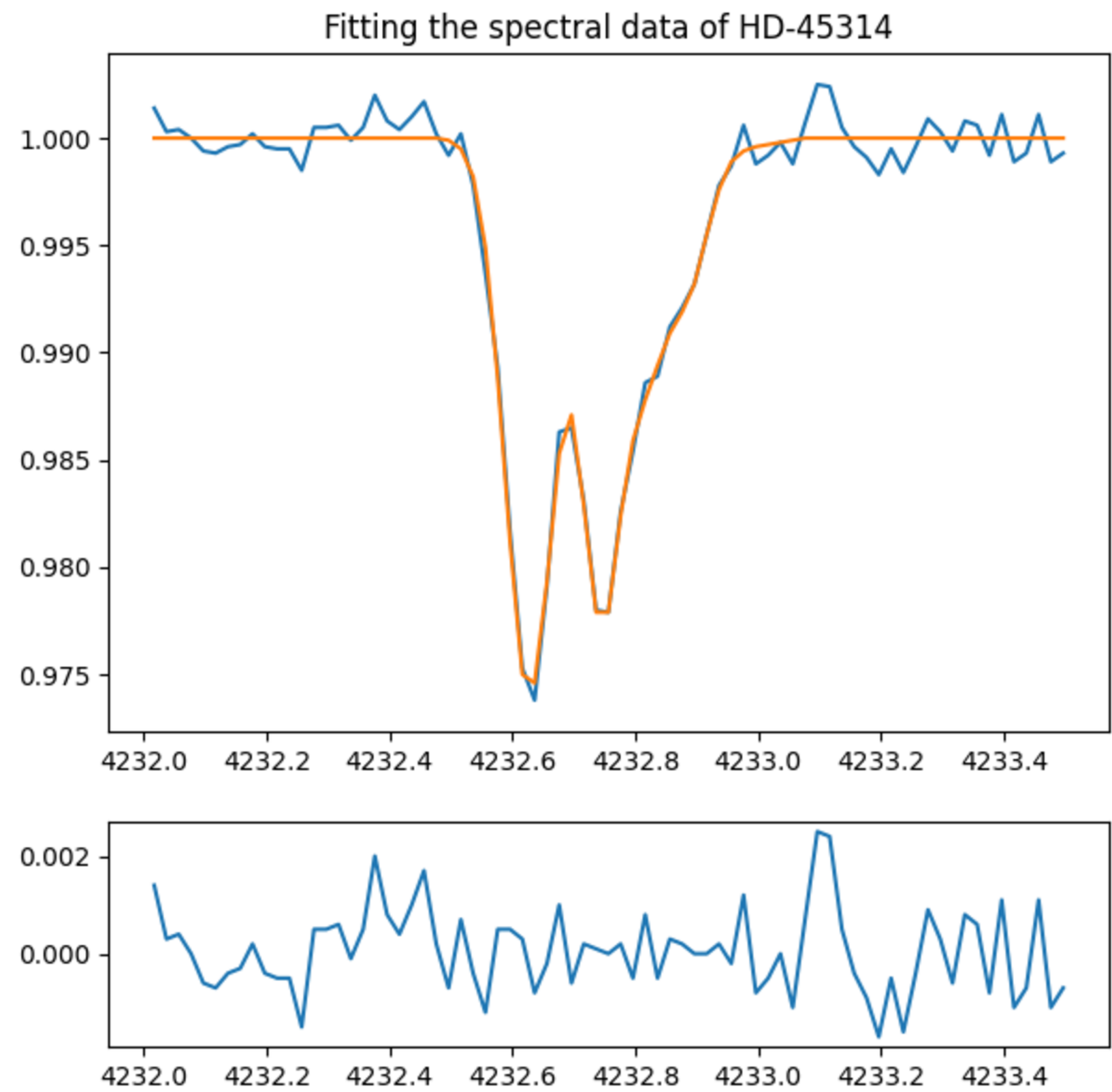


Sightline	N	b	v
HD186745	$1.5^{0.1}_{0.1}$	3.6 ± 0.1	-10.4 ± 0.1
	$1.6^{0.1}_{0.1}$	3.3 ± 0.1	-2.6 ± 0.1



Sightline	N	b	v	R	N _{lit}	b _{lit}	v _{lit}	R _{lit}	Reference
HD75860	$2.4^{0.4}_{0.3}$	3.6 ± 0.2	23.4 ± 0.2	51^{26}_{18}	7.5 ± 0.5	..	21.6	..	1
	$4.0^{0.4}_{0.3}$	3.3 ± 0.2	19.2 ± 0.3	63^{25}_{18}					

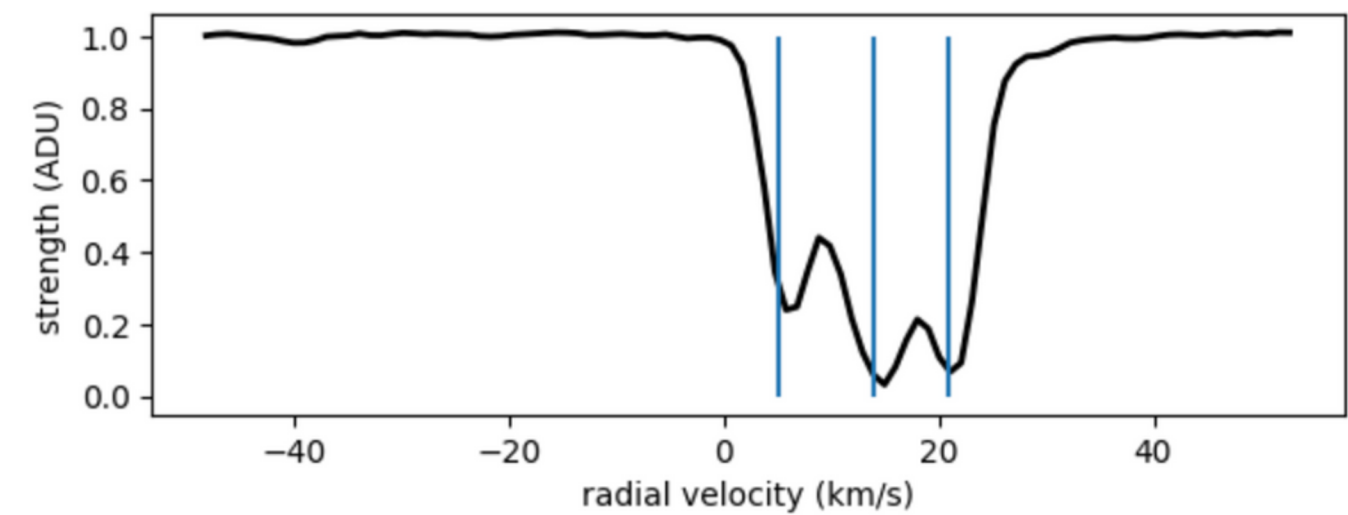
A small subset of the results (triple cloud models)



Na-I lines for comparison

Sightline	N	b	v
HD45314	$0.21^{0.06}_{0.05}$	6.3 ± 1.3	19.6 ± 1.4
	$0.16^{0.06}_{0.04}$	2.5 ± 0.5	13.8 ± 0.2
	$0.30^{0.01}_{0.01}$	3.5 ± 0.2	5.7 ± 0.1

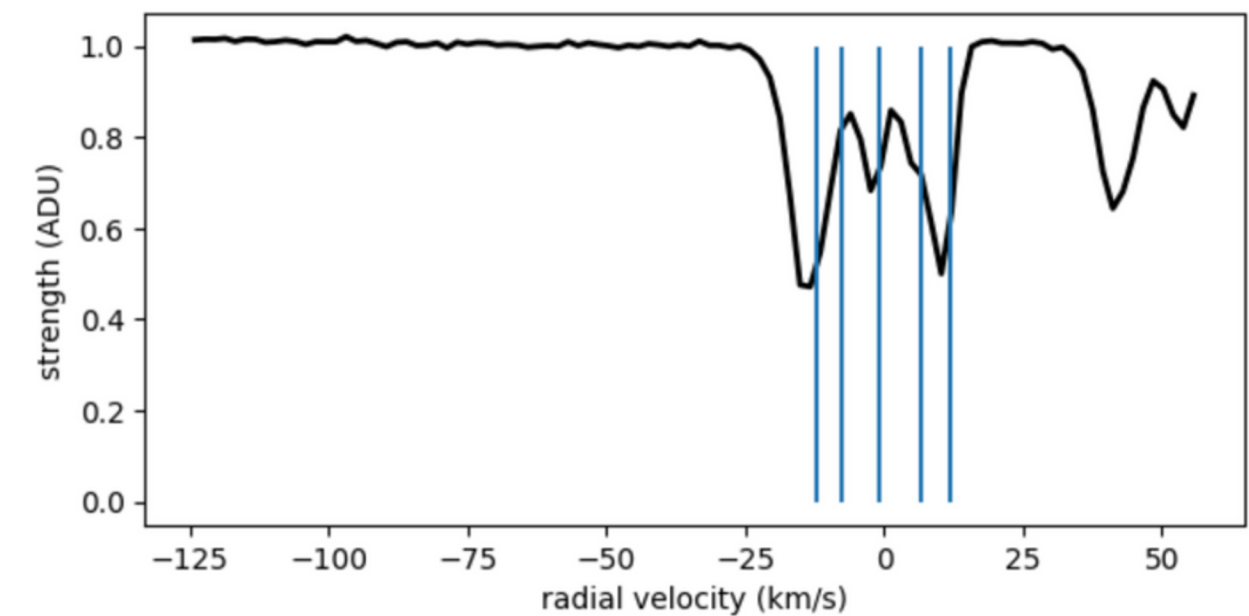
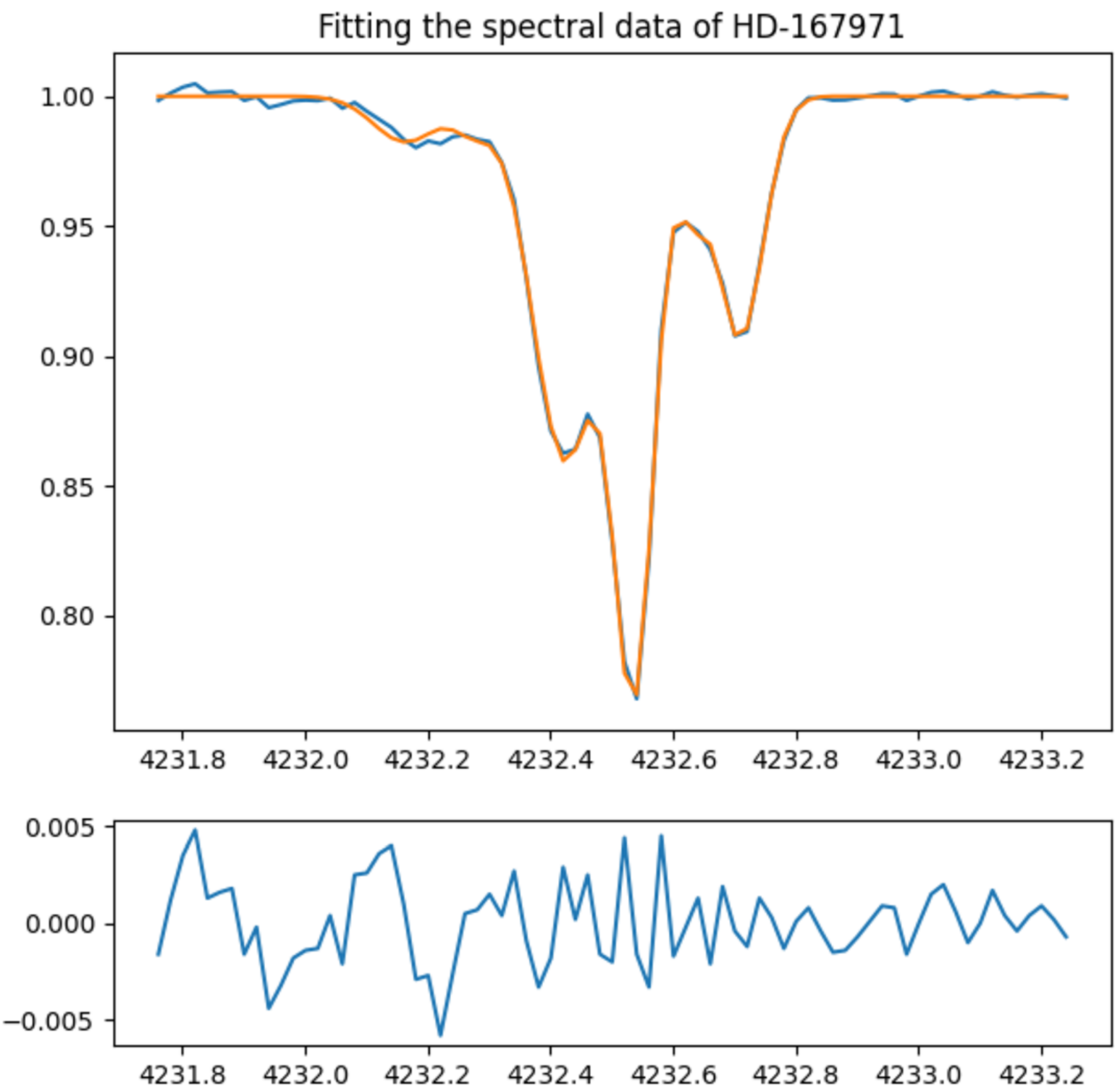
A small subset of the results (triple cloud models)



Na-II lines for comparison

Sightline	N	b	v
	$0.04^{0.01}_{0.01}$	1.8 ± 0.9	4.4 ± 0.4
HD40111	$0.26^{0.02}_{0.02}$	3.1 ± 0.3	14.3 ± 0.2
	$0.11^{0.02}_{0.01}$	1.9 ± 0.4	20.7 ± 0.2

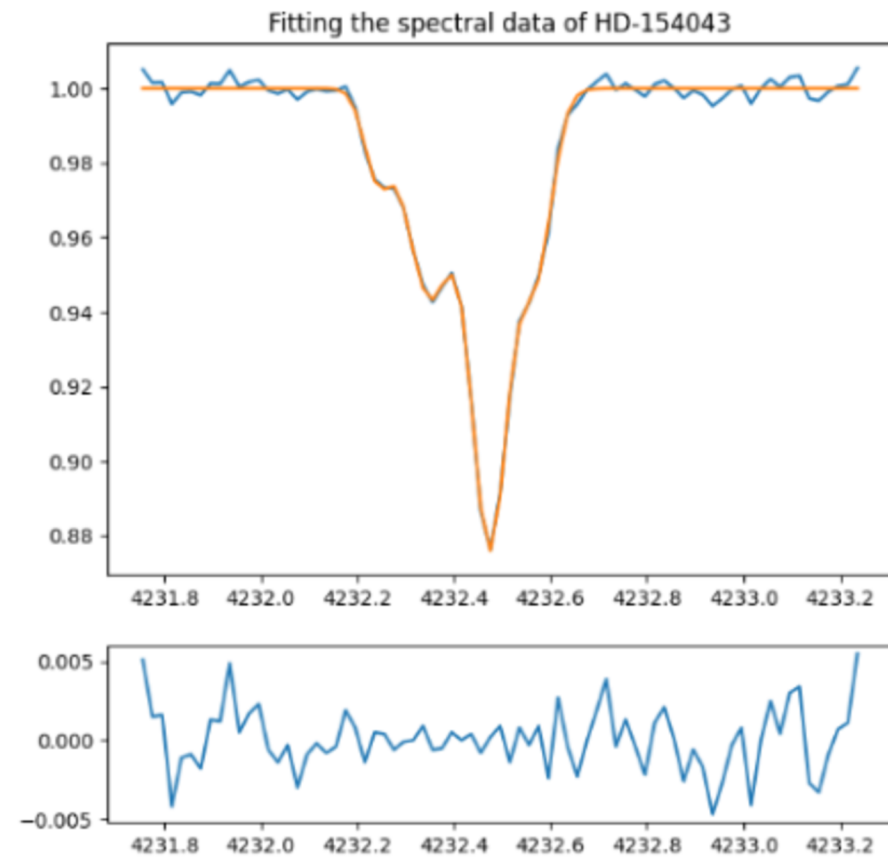
A small subset of the results (four cloud models)



Na-I lines for comparison

Sightline	N	b	v	R
HD167971	$1.1^{0.1}_{0.1}$	3.3 ± 0.2	11.4 ± 0.1	..
	$2.4^{0.1}_{0.1}$	2.6 ± 0.1	-0.8 ± 0.1	20^4_3
	$2.3^{0.3}_{0.3}$	4.8 ± 0.2	-8.8 ± 0.4	8^2_1
	$0.25^{0.05}_{0.05}$	1.2 ± 0.6	5.8 ± 0.2	..

A small subset of the results (four cloud models)



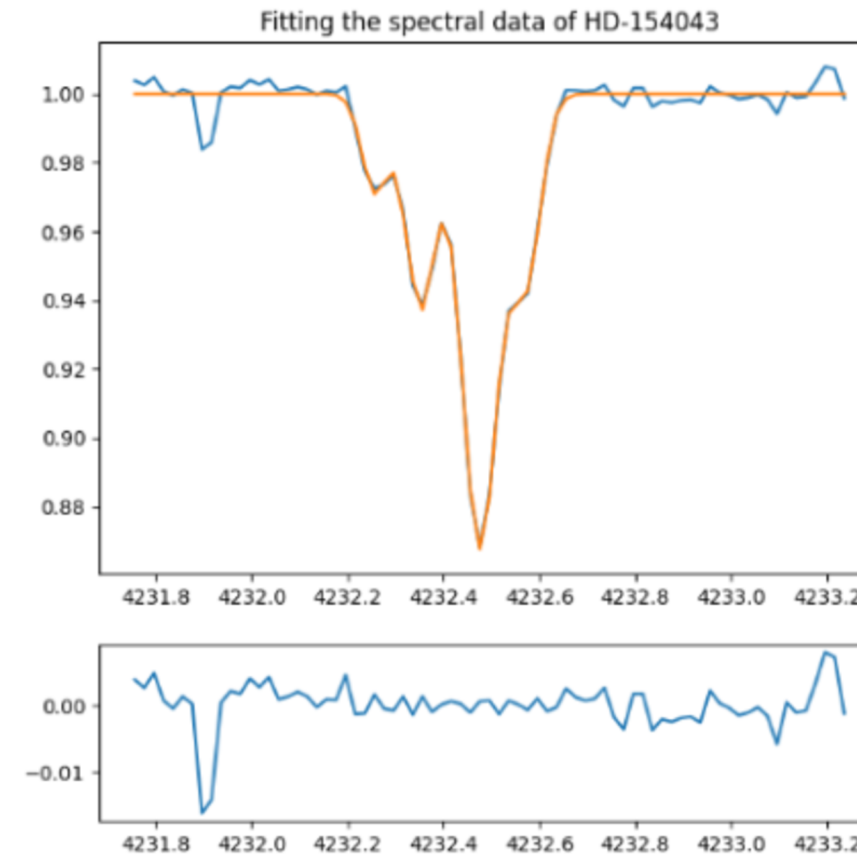
Best fit parameters:

■ ■ ■ ■ b ■ ■ ■ log(N)

12CHI 1		
$v = +1.5 \pm 0.3$	2.80 ± 0.33	12.6941 ± 0.0494
$v = -5.2 \pm 0.1$	3.22 ± 0.25	13.1496 ± 0.0290
$v = -14.0 \pm 0.2$	4.35 ± 0.56	12.8991 ± 0.0426
$v = -21.7 \pm 0.3$	1.93 ± 0.52	12.2531 ± 0.0834

13CHI 2

- Saved the fitted lines to PDF file: HD 154043_plot.pdf
chi-squared value 0.3368893173868607



Best fit parameters:

■ ■ ■ ■ b ■ ■ ■ log(N)

12CHI 1		
$v = +1.7 \pm 0.3$	2.44 ± 0.40	12.7048 ± 0.0516
$v = -5.1 \pm 0.1$	3.12 ± 0.23	13.1781 ± 0.0209
$v = -13.8 \pm 0.2$	2.54 ± 0.38	12.7671 ± 0.0359
$v = -20.6 \pm 0.3$	1.90 ± 0.61	12.3592 ± 0.0636

13CHI 2

- Saved the fitted lines to PDF file: HD 154043_plot.pdf
chi-squared value 0.903574219702956



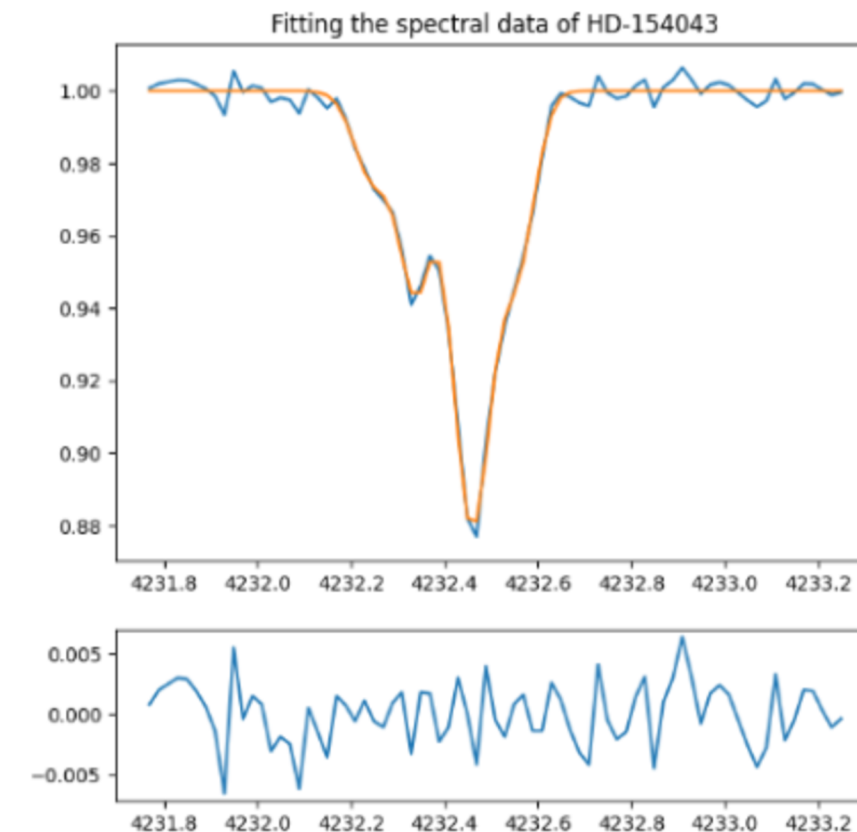
Best fit parameters:

■ ■ ■ ■ b ■ ■ ■ log(N)

12CHI 1		
$v = +1.2 \pm 0.4$	2.72 ± 0.48	12.7078 ± 0.0613
$v = -5.6 \pm 0.1$	2.99 ± 0.26	13.1674 ± 0.0264
$v = -14.2 \pm 0.2$	3.03 ± 0.52	12.7883 ± 0.0458
$v = -21.0 \pm 0.4$	1.92 ± 0.69	12.3538 ± 0.0845

13CHI 2

- Saved the fitted lines to PDF file: HD 154043_plot.pdf
chi-squared value 0.9016316421218195



Best fit parameters:

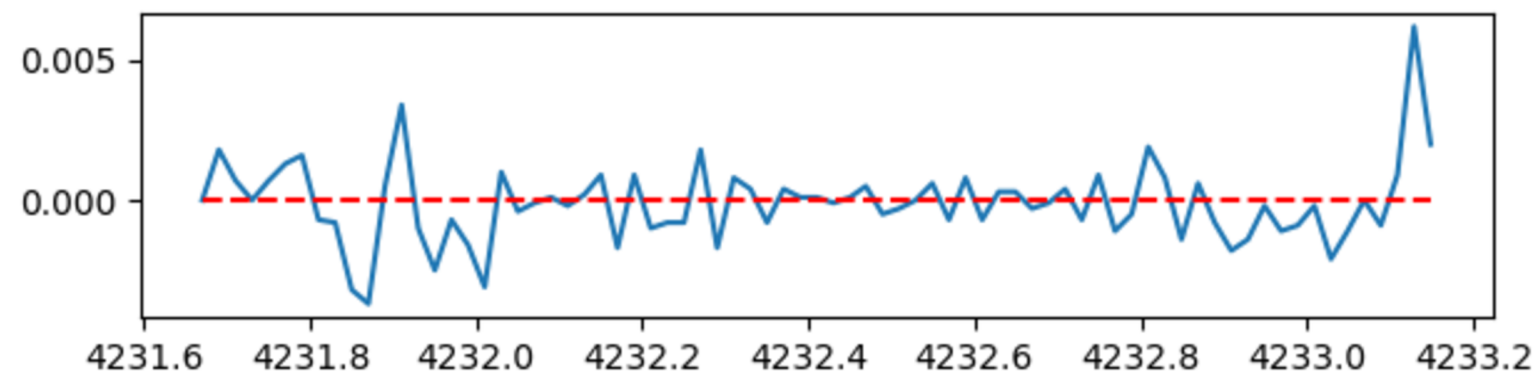
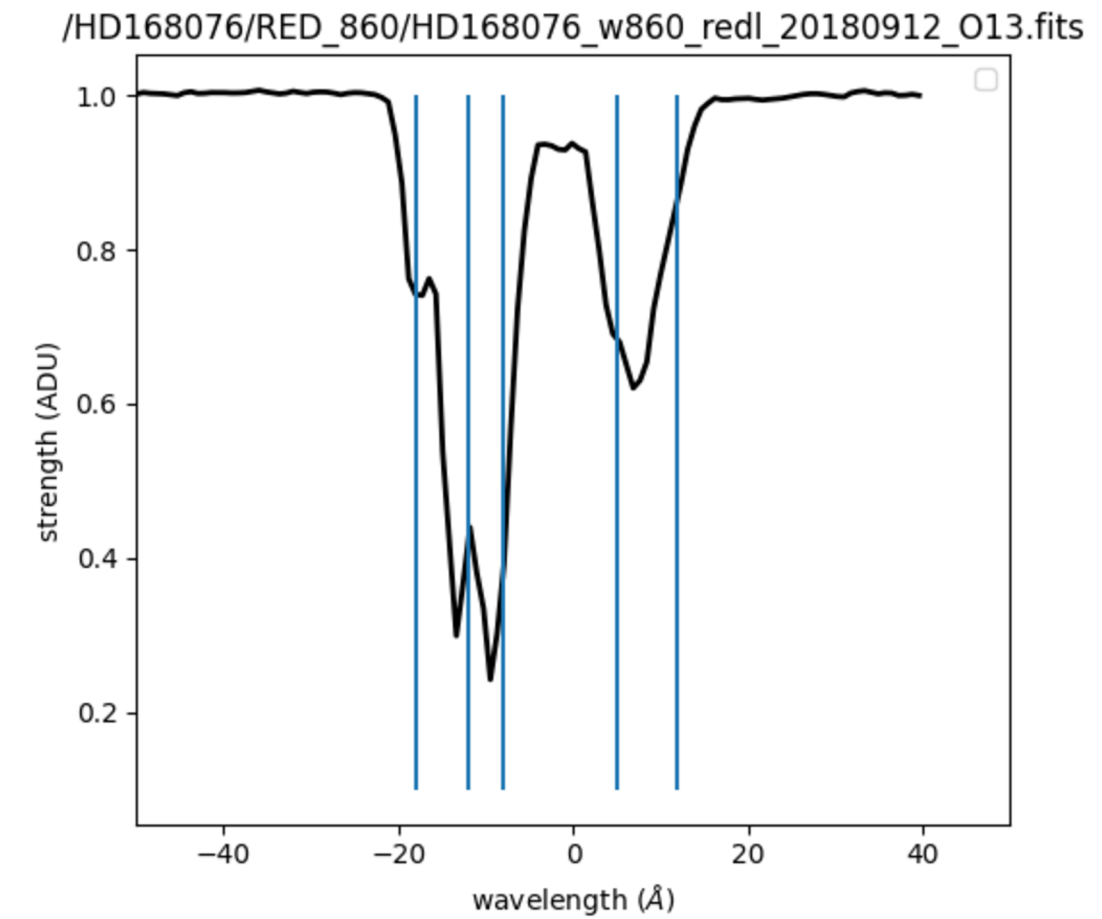
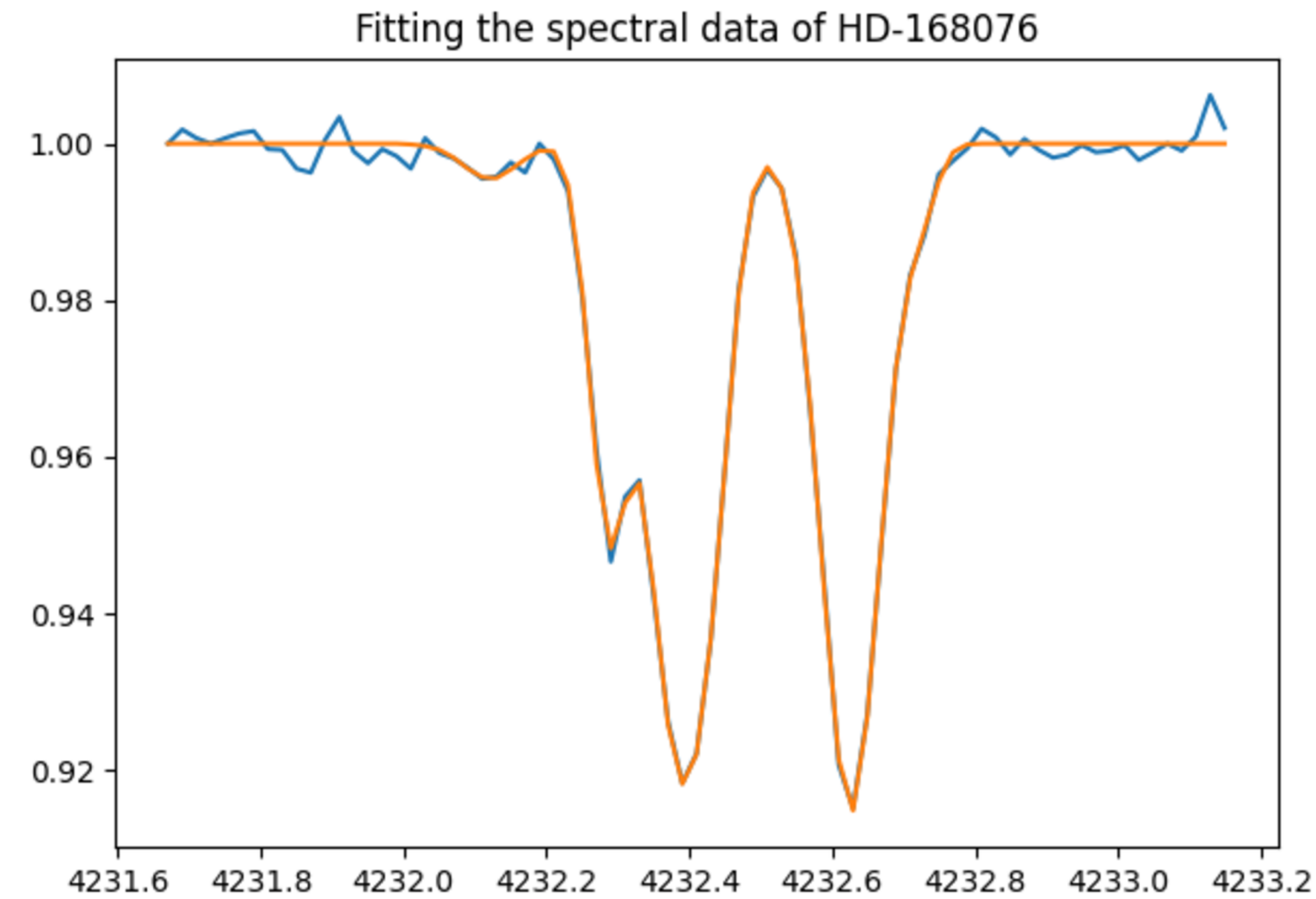
■ ■ ■ ■ b ■ ■ ■ log(N)

12CHI 1		
$v = +0.7 \pm 0.5$	3.11 ± 0.51	12.6966 ± 0.0837
$v = -6.3 \pm 0.2$	3.64 ± 0.36	13.1852 ± 0.0357
$v = -14.8 \pm 0.4$	2.80 ± 0.67	12.7324 ± 0.1160
$v = -21.1 \pm 1.2$	3.66 ± 1.23	12.4803 ± 0.1616

13CHI 2

- Saved the fitted lines to PDF file: HD 154043_plot.pdf
chi-squared value 0.5653829593668918

A small subset of the results (five cloud models)



■ ■ ■ ■ b ■ ■ ■ log(N)

12CHI 1

$v = -11.9 \pm 0.6$

3.32 ± 0.68

12.9327 ± 0.1177

$v = -8.1 \pm 0.6$

2.19 ± 0.51

12.4334 ± 0.3195

$v = -18.5 \pm 0.1$

1.82 ± 0.22

12.5751 ± 0.0394

$v = +5.6 \pm 0.1$

3.66 ± 0.10

13.0280 ± 0.0065

$v = +12.6 \pm 0.5$

0.97 ± 1.36

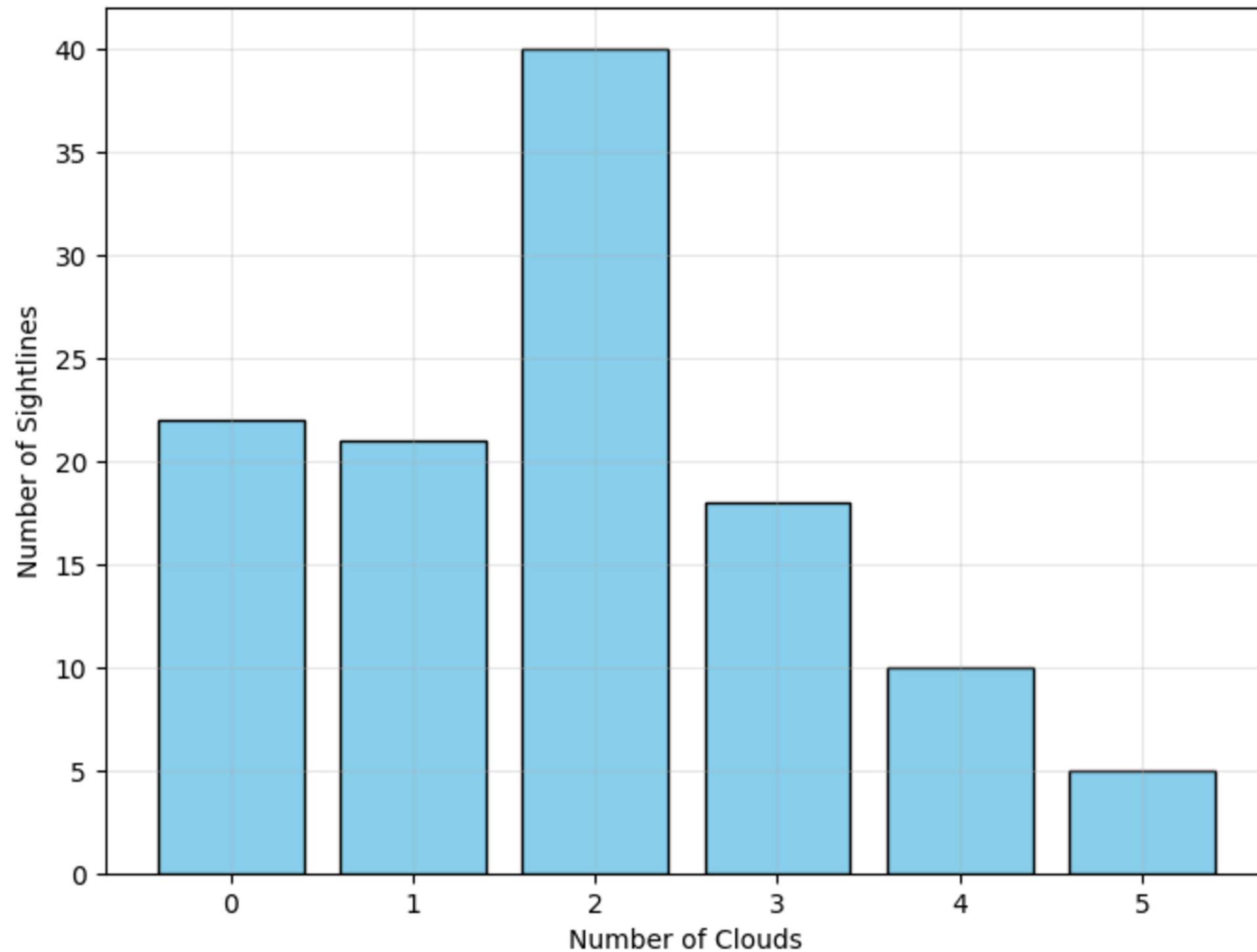
11.6552 ± 0.1302

13CHI 2

$v = -11.9 \pm 0.6$

3.32 ± 0.68

11.6958 ± 0.0914



- Apart from reduced chi-squared value AIC/BIC is used to judge the fitting and optimum number of components.
- CH⁺ is very abundant.
- 13CH⁺ signatures although visible, are hard to quantify. The strongest single cloud sightlines point towards a 12CH⁺/13CH⁺ ratio of ~50-80 but it varies a lot.